

Prvotná štatistická analýza, EDA

Jana Kalická

2026-09-25

Užitocne prikazy na odhalenie struktury dat

```
diabetes <- read.csv("diabetes.csv")
str(diabetes)

## 'data.frame': 768 obs. of 9 variables:
## $ Pregnancies : int 6 1 8 1 0 5 3 10 2 8 ...
## $ Glucose : int 148 85 183 89 137 116 78 115 197 125 ...
## $ BloodPressure : int 72 66 64 66 40 74 50 0 70 96 ...
## $ SkinThickness : int 35 29 0 23 35 0 32 0 45 0 ...
## $ Insulin : int 0 0 0 94 168 0 88 0 543 0 ...
## $ BMI : num 33.6 26.6 23.3 28.1 43.1 25.6 31 35.3 30.
5 0 ...
## $ DiabetesPedigreeFunction: num 0.627 0.351 0.672 0.167 2.288 ...
## $ Age : int 50 31 32 21 33 30 26 29 53 54 ...
## $ Outcome : int 1 0 1 0 1 0 1 0 1 1 ...

names(diabetes)

## [1] "Pregnancies" "Glucose"
## [3] "BloodPressure" "SkinThickness"
## [5] "Insulin" "BMI"
## [7] "DiabetesPedigreeFunction" "Age"
## [9] "Outcome"

head(diabetes)

## Pregnancies Glucose BloodPressure SkinThickness Insulin BMI
## 1 6 148 72 35 0 33.6
## 2 1 85 66 29 0 26.6
## 3 8 183 64 0 0 23.3
## 4 1 89 66 23 94 28.1
## 5 0 137 40 35 168 43.1
## 6 5 116 74 0 0 25.6
## DiabetesPedigreeFunction Age Outcome
## 1 0.627 50 1
## 2 0.351 31 0
## 3 0.672 32 1
## 4 0.167 21 0
## 5 2.288 33 1
## 6 0.201 30 0
```

ked casto pouzivam kniznicu, tak nadcitam

```
library(readxl)
```

ak potrebujem iba jeden prikaz kniznice, staci takto

```
data <- readxl::read_xlsx("data_vyuka.xlsx")
str(data)

## tibble [50 × 13] (S3: tbl_df/tbl/data.frame)
## $ respondent: num [1:50] 1 2 3 4 5 6 7 8 9 10 ...
## $ pohlavie  : chr [1:50] "m" "m" "m" "z" ...
## $ vek       : num [1:50] 28 22 50 39 53 55 50 39 57 45 ...
## $ vzdelanie : num [1:50] 3 2 1 2 3 1 3 1 2 2 ...
## $ stav      : num [1:50] 1 1 2 2 2 3 2 2 2 2 ...
## $ fajciar   : num [1:50] 0 0 0 0 0 0 0 1 0 0 ...
## $ praca     : num [1:50] 2 1 2 3 1 3 3 3 1 2 ...
## $ clen      : num [1:50] 5 4 4 5 5 2 3 4 2 5 ...
## $ mprij     : num [1:50] 1500 900 975 1200 1800 ...
## $ mstrava   : num [1:50] 250 200 350 400 400 200 350 250 300 320 ...
## $ rdovolenka: num [1:50] 10000 350 300 500 15000 200 900 200 250 450 ...
## $ pocit     : num [1:50] 73 27 65 26 38 19 42 57 10 62 ...
## $ invest    : num [1:50] 2 4 2 3 2 4 2 3 4 1 ...

head(data)

## # A tibble: 6 × 13
##   respondent pohlavie   vek vzdelanie   stav fajciar praca   clen mprij mstr
ava
##       <dbl> <chr>      <dbl>      <dbl> <dbl>   <dbl> <dbl> <dbl> <dbl> <d
bl>
## 1           1 m          28          3      1       0      2      5 1500
250
## 2           2 m          22          2      1       0      1      4  900
200
## 3           3 m          50          1      2       0      2      4  975
350
## 4           4 z          39          2      2       0      3      5 1200
400
## 5           5 z          53          3      2       0      1      5 1800
400
## 6           6 z          55          1      3       0      3      2  825
200
## # i 3 more variables: rdovolenka <dbl>, pocit <dbl>, invest <dbl>

View(data)
```

Chybajuce data

Kniznice mice, Amelia

```
is.na(data)#nevhodny vystup
```

```
##      respondent pohlavie vek vzdelanie stav fajciar praca clen mprij
## [1,]      FALSE      FALSE FALSE      FALSE FALSE      FALSE FALSE FALSE FALSE
## [2,]      FALSE      FALSE FALSE      FALSE FALSE      FALSE FALSE FALSE FALSE
## [3,]      FALSE      FALSE FALSE      FALSE FALSE      FALSE FALSE FALSE FALSE
## [4,]      FALSE      FALSE FALSE      FALSE FALSE      FALSE FALSE FALSE FALSE
FALSE FALSE FALSE      FALSE FALSE      FALSE FALSE FALSE FALSE
##      mstrava rdovolenka pocit invest
## [1,]      FALSE      FALSE FALSE      FALSE
## [2,]      FALSE      FALSE FALSE      FALSE
## [3,]      FALSE      FALSE FALSE      FALSE
## [4,]      FALSE      FALSE FALSE      FALSE
```

kolko ich napr v stlpci vzdelanie

```
sum(is.na(data$vzdelanie))

## [1] 2

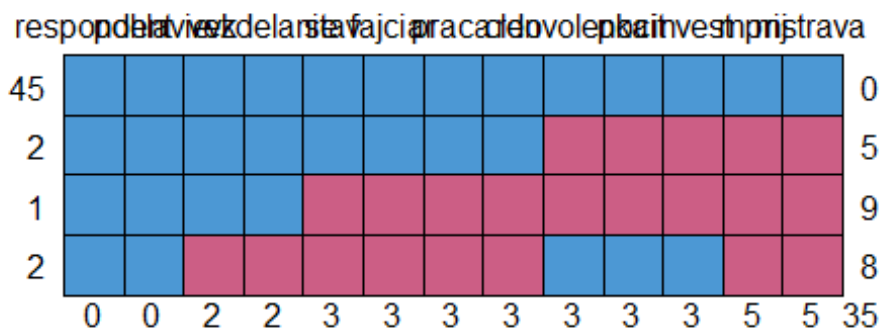
library(mice)

##
## Attaching package: 'mice'

## The following object is masked from 'package:stats':
##
##      filter

## The following objects are masked from 'package:base':
##
##      cbind, rbind

md.pattern(data)# vizualizacia chybyajucich a nechbyajucich dat
```



```
## respondent pohlavie vek vzdelanie stav fajciar praca clen rdovolenka po
cit
## 45      1      1  1      1  1      1  1  1      1
1
## 2      1      1  1      1  1      1  1  1      0
0
## 1      1      1  1      1  0      0  0  0      0
0
## 2      1      1  0      0  0      0  0  0      1
1
##      0      0  2      2  3      3  3  3      3
3
## invest mprij mstrava
## 45      1      1      1  0
## 2      0      0      0  5
## 1      0      0      0  9
## 2      1      0      0  8
##      3      5      5 35
```

```
library(Amelia)
```

```
## Loading required package: Rcpp
```

```
## ##
```

```
## ## Amelia II: Multiple Imputation
```

```
## ## (Version 1.8.3, built: 2024-11-07)
```

```
## ## Copyright (C) 2005-2026 James Honaker, Gary King and Matthew Blackwell
```

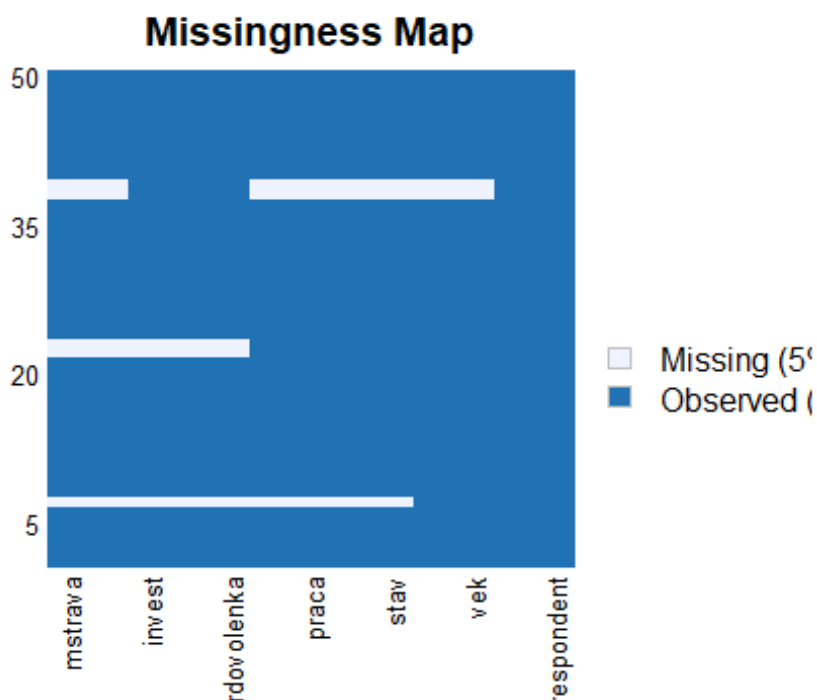
```
## ## Refer to http://gking.harvard.edu/amelia/ for more information
## ##
```

```
missmap(data)#iny typ vizualizacie
```

```
## Warning: Unknown or uninitialised column: `arguments`.
```

```
## Unknown or uninitialised column: `arguments`.
```

```
## Warning: Unknown or uninitialised column: `imputations`.
```



Kedze chybajuce data nedoplname, prislusne riadky vynechame a pracujeme s uplnym datasetom

```
new_data <- na.omit(data)
str(new_data)
```

```
## tibble [45 × 13] (S3: tbl_df/tbl/data.frame)
## $ respondent: num [1:45] 1 2 3 4 5 6 7 8 9 10 ...
## $ pohlavie : chr [1:45] "m" "m" "m" "z" ...
## $ vek : num [1:45] 28 22 50 39 53 55 50 39 57 45 ...
## $ vzdelanie : num [1:45] 3 2 1 2 3 1 3 1 2 2 ...
## $ stav : num [1:45] 1 1 2 2 2 3 2 2 2 2 ...
## $ fajciar : num [1:45] 0 0 0 0 0 0 0 1 0 0 ...
## $ praca : num [1:45] 2 1 2 3 1 3 3 3 1 2 ...
## $ clen : num [1:45] 5 4 4 5 5 2 3 4 2 5 ...
## $ mprij : num [1:45] 1500 900 975 1200 1800 ...
## $ mstrava : num [1:45] 250 200 350 400 400 200 350 250 300 320 ...
## $ rdovolenka: num [1:45] 10000 350 300 500 15000 200 900 200 250 450 ...
```

```
## $ pocit      : num [1:45] 73 27 65 26 38 19 42 57 10 62 ...
## $ invest     : num [1:45] 2 4 2 3 2 4 2 3 4 1 ...
## - attr(*, "na.action")= 'omit' Named int [1:5] 12 13 28 29 44
## ..- attr(*, "names")= chr [1:5] "12" "13" "28" "29" ...
```

Vhodne kniznice pre EDA

```
library(psych)
library(Hmisc)

##
## Attaching package: 'Hmisc'

## The following object is masked from 'package:psych':
##
##     describe

## The following objects are masked from 'package:base':
##
##     format.pval, units

library(FSA)

## ## FSA v0.10.1. See citation('FSA') if used in publication.
## ## Run fishR() for related website and fishR('IFAR') for related book.

##
## Attaching package: 'FSA'

## The following object is masked from 'package:psych':
##
##     headtail

library(pastecs)
library(moments)
library(mosaic)

## Registered S3 method overwritten by 'mosaic':
##   method                        from
##   fortify.SpatialPolygonsDataFrame ggplot2

##
## The 'mosaic' package masks several functions from core packages in order to add
## additional features. The original behavior of these functions should not
## be affected by this.

##
## Attaching package: 'mosaic'
```

```
## The following objects are masked from 'package:dplyr':
##
##   count, do, tally

## The following object is masked from 'package:Matrix':
##
##   mean

## The following object is masked from 'package:ggplot2':
##
##   stat

## The following object is masked from 'package:FSA':
##
##   perc

## The following objects are masked from 'package:psych':
##
##   logit, rescale

## The following objects are masked from 'package:stats':
##
##   binom.test, cor, cor.test, cov, fivenum, IQR, median, prop.test,
##   quantile, sd, t.test, var

## The following objects are masked from 'package:base':
##
##   max, mean, min, prod, range, sample, sum

library(ie2misc)# kvoli MAD
library(modeest)# modus

## Registered S3 method overwritten by 'rmutil':
##   method          from
##   plot.residuals psych

##
## Attaching package: 'modeest'

## The following object is masked from 'package:moments':
##
##   skewness
```

Charakteristiky polohy

Urcime charakteristiky polohy pre stlpec mprij, novy nazov plat

```
plat <- new_data$mprij
```

Tabulky pocetnosti

```
table(plat)#tabulka pocetnosti
```

```
## plat
## 600 675 750 825 900 975 1050 1125 1200 1275 1350 1425 1500 1650 1800
1950
## 1 2 3 5 5 5 3 2 5 3 4 1 1 2 2
1

prop.table(table(plat))#relativne pocetnosti

## plat
## 600 675 750 825 900 975 10
50
## 0.02222222 0.04444444 0.06666667 0.11111111 0.11111111 0.11111111 0.066666
67
## 1125 1200 1275 1350 1425 1500 16
50
## 0.04444444 0.11111111 0.06666667 0.08888889 0.02222222 0.02222222 0.044444
44
## 1800 1950
## 0.04444444 0.02222222

cumsum(table(plat))#kumulativne pocetnosti

## 600 675 750 825 900 975 1050 1125 1200 1275 1350 1425 1500 1650 1800
1950
## 1 3 6 11 16 21 24 26 31 34 38 39 40 42 44
45
```

Tabulka triednych pocetnosti, viacerymi prikazmi urcime hranice a pocatnosti pomocou prikazu hist

```
hist(plat,breaks = 3,plot=F)

## $breaks
## [1] 500 1000 1500 2000
##
## $counts
## [1] 21 19 5
##
## $density
## [1] 0.0009333333 0.0008444444 0.0002222222
##
## $mids
## [1] 750 1250 1750
##
## $xname
## [1] "plat"
##
## $equidist
## [1] TRUE
##
```

```
## attr(,"class")
## [1] "histogram"

tp <-hist(plat,breaks = 3,plot=F)
tp$breaks

## [1] 500 1000 1500 2000

tp$counts

## [1] 21 19 5
```

Charakteristiky polohy

Najprv cez jediny prikazy, potom aj sumarne

```
min(new_data$mprij)

## [1] 600

max(new_data$mprij)

## [1] 1950

mean(new_data$mprij)#aritmeticky priemer, ale nemam chybajuce hodnoty

## [1] 1113.333

mean(data$mprij,na.rm=T)# chybajuce sa vynechaju a zrata sa statistika

## [1] 1113.333

median(plat)#median

## [1] 1050

mfv(plat)#modus

## [1] 825 900 975 1200

quantile(plat) #5 hodnot

## 0% 25% 50% 75% 100%
## 600 900 1050 1275 1950

fivenum(plat)

## [1] 600 900 1050 1275 1950

quantile(plat,0.1) # 0.1 kvantil

## 10%
## 750
```

Uvediem prikazy prikazy pre harmonicky a geometricky priemer

```
harmonic.mean(plat)
## [1] 1029.843
geometric.mean(plat)
## [1] 1070.179
```

Uziticne prikazy

```
sort(plat)
## [1] 600 675 675 750 750 750 825 825 825 825 825 900 900 900
## [16] 900 975 975 975 975 975 1050 1050 1050 1125 1125 1200 1200 1200
## [31] 1200 1275 1275 1275 1350 1350 1350 1350 1425 1500 1650 1650 1800 1800
## [46] 1950

sort(plat)[15]# hodnota v poradi 15
## [1] 900

sum(plat==1050)# pocet respondentov s platom 1050
## [1] 3

sum(new_data$vek<40)
## [1] 22
```

Charakteristiky rozptylu

variace rozpatie, odhad disperzie, smerodajna odchylka kvartilove rozpatie, IQR

```
max(plat)-min(plat)
## [1] 1350

range(plat)
## [1] 600 1950

var(plat)#disperzia
## [1] 104562.5

sd(plat)#smerodajna odchylka
## [1] 323.3613
```

```

mad(plat)#medianova absolutna odchylka
## [1] 333.585

madstat(plat)#priemerna absolutna
## [1] 260.8889

IQR(plat)
## [1] 375

library(DescTools)

## Registered S3 method overwritten by 'httr':
##   method      from
##   print.response rutil

##
## Attaching package: 'DescTools'

## The following object is masked from 'package:mosaic':
##
##   MAD

## The following objects are masked from 'package:Hmisc':
##
##   %nin%, Label, Mean, Quantile

## The following objects are masked from 'package:psych':
##
##   AUC, ICC, SD

CoefVar(plat)#variacyjny koeficient
## [1] 0.2904442

```

Ine charakteristiky

Pre sikmost a spicatost preferovana kniznica moments

```

skew(plat)#sikmost, nie tato kniznica
## [1] 0.7028482

moments::skewness(plat)
## [1] 0.7269445

kurtosi(plat)# nepouzivat
## [1] -0.1756536

```

```
kurtosis(plat)
```

```
## [1] 2.954185
```

Simulacie symetrickych a nesymetrickych rozdeleni a ich sikmosti

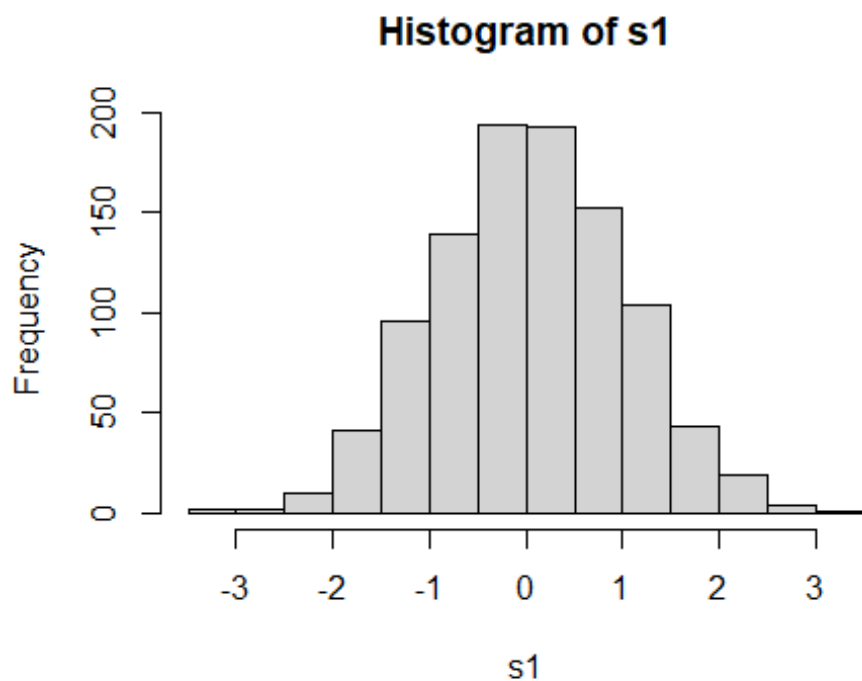
Normalne rozdelenie data

```
s1 <- rnorm(1000,mean=0,sd=1)
```

```
moments::skewness((s1))
```

```
## [1] 0.002482011
```

```
hist(s1)
```



```
kurtosis(s1)
```

```
## [1] 2.974197
```

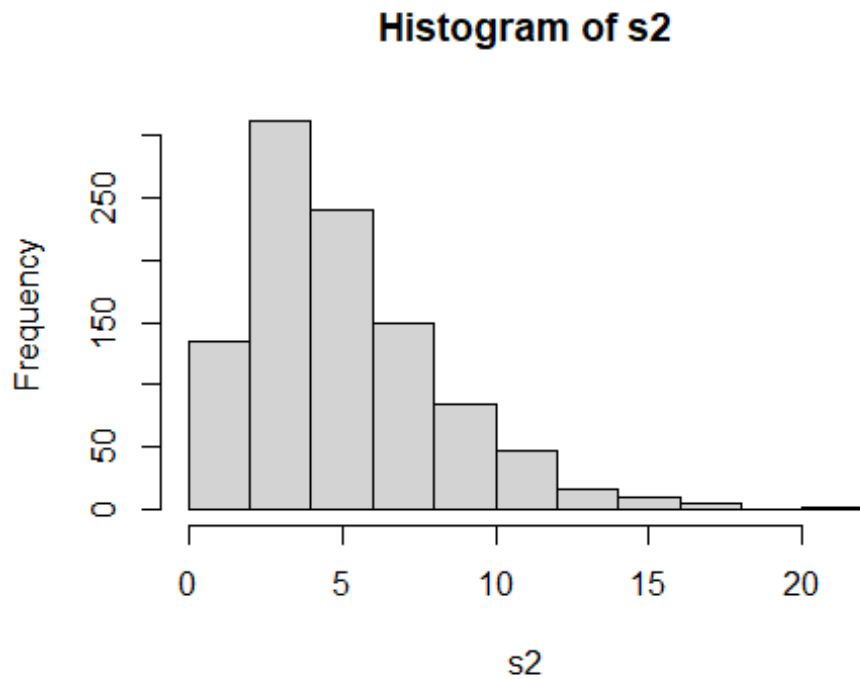
Chi kvadrat rozdelenie data

```
s2 <- rchisq(1000,5)
```

```
moments::skewness((s2))
```

```
## [1] 1.108971
```

```
hist(s2)
```



```
kurtosis(s2)
```

```
## [1] 4.395693
```

Sumarne statistiky

```
describe(plat)
```

```
## plat
##      n missing distinct      Info      Mean  pMedian      Gmd      .05
##      45       0       16    0.993    1113    1088    363    690
##      .10     .25     .50     .75     .90     .95
##      750     900    1050    1275    1590    1770
##
## Value      600    675    750    825    900    975    1050    1125    1200    1275    13
50
## Frequency      1      2      3      5      5      5      3      2      5      3
4
## Proportion 0.022 0.044 0.067 0.111 0.111 0.111 0.067 0.044 0.111 0.067 0.0
89
##
## Value      1425    1500    1650    1800    1950
## Frequency      1      1      2      2      1
## Proportion 0.022 0.022 0.044 0.044 0.022
```

```
round(stat.desc(plat),2)
```

```
##      nbr.val      nbr.null      nbr.na      min      max      ra
nge
##      45.00      0.00      0.00      600.00      1950.00      1350.
00
##      sum      median      mean      SE.mean CI.mean.0.95
var
##      50100.00      1050.00      1113.33      48.20      97.15      104562.
50
##      std.dev      coef.var
##      323.36      0.29
```

```
summary(plat)
```

```
##      Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
##      600      900      1050      1113      1275      1950
```

Charakteristiky pre kazdy stlpec tabulky

```
inspect(new_data)
```

```
##
## categorical variables:
##      name      class levels  n missing
## 1 pohlavie character      2 45      0
##                                distribution
## 1 z (51.1%), m (48.9%)
##
## quantitative variables:
##      name      class min  Q1 median  Q3  max      mean      sd
n
## 1 respondent numeric    1  14    25   38   50  25.5333333  14.8578109
45
## 2      vek numeric    20  33    42   53   60  42.2000000  12.1871766
45
## 3  vzdelanie numeric     1   1     2    3    3   2.0000000   0.7385489
45
## 4      stav numeric     1   2     2    2    4   2.0444444   0.5622825
45
## 5   fajciar numeric     0   0     0    1    1   0.2666667   0.4472136
45
## 6      praca numeric     1   1     2    3    5   2.3111111   1.2214414
45
## 7      clen numeric     1   2     3    4    5   3.1555556   1.2784144
45
## 8      mprij numeric   600 900   1050 1275 1950 1113.3333333 323.3612531
45
## 9   mstrava numeric   100 240    260  350   400  279.5555556   72.7997114
45
```

```
## 10 rdovolenka numeric 100 250 350 900 15000 2650.0000000 4648.5090474
45
## 11 pocit numeric 10 36 50 65 95 49.8000000 20.7381949
45
## 12 invest numeric 1 2 2 4 5 2.6222222 1.2300940
45
## missing
## 1 0
## 2 0
## 3 0
## 4 0
## 5 0
## 6 0
## 7 0
## 8 0
## 9 0
## 10 0
## 11 0
## 12 0
```

```
library(skimr)
```

```
##
```

```
## Attaching package: 'skimr'
```

```
## The following object is masked from 'package:mosaic':
```

```
##
```

```
## n_missing
```

```
skim(new_data)
```

Data summary

Name	new_data
Number of rows	45
Number of columns	13

Column type frequency:

character	1
numeric	12

Group variables	None
-----------------	------

Variable type: character

skim_variable	n_missing	complete_rate	mi n	m ax	empt y	n_unique	whitespace
---------------	-----------	---------------	---------	---------	-----------	----------	------------

skim_variable	n_missing	complete_rate	mi n	m ax	empt y	n_unique	whitespace
pohlavie	0	1	1	1	0	2	0

Variable type: numeric

skim_variabl e	n_missin g	complete_rat e	mean	sd	p 0	p 2 5	p5 0	p7 5	p10 0	hist
respondent	0	1	25.53	14.86	1	1 4	25	38	50	
vek	0	1	42.20	12.19	2 0	3 3	42	53	60	
vzdelanie	0	1	2.00	0.74	1	1	2	3	3	
stav	0	1	2.04	0.56	1	2	2	2	4	
fajciar	0	1	0.27	0.45	0	0	0	1	1	
praca	0	1	2.31	1.22	1	1	2	3	5	
clen	0	1	3.16	1.28	1	2	3	4	5	
mprij	0	1	1113. 33	323.3 6	6 0 0	9 0 0	10 50	12 75	195 0	
mstrava	0	1	279.5 6	72.80	1 0 0	2 4 0	26 0	35 0	400	
rdovolenka	0	1	2650. 00	4648. 51	1 0 0	2 5 0	35 0	90 0	150 00	

skim_variabl	n_missin	complete_rat				p	p	p5	p7	p10	
e	g	e	mean	sd	0	2	5	0	5	0	hist
pocit	0	1	49.80	20.74	1	3	50	65	95		
					0	6					
invest	0	1	2.62	1.23	1	2	2	4	5		
											

Charakteristiky na vybranych podmnozinach

muzi a zeny

```
tapply(new_data$mprij,new_data$pohlavie,mean)

##           m           z
## 1067.045 1157.609

tapply(new_data$mprij,new_data$pohlavie,summary)

## $m
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   675.0   843.8   975.0  1067.0  1200.0  1950.0
##
## $z
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    600    900   1125   1158   1350   1800
```

Grafy EDA

Vhodne kniznice

```
library(ggplot2)
library(car)

## Loading required package: carData

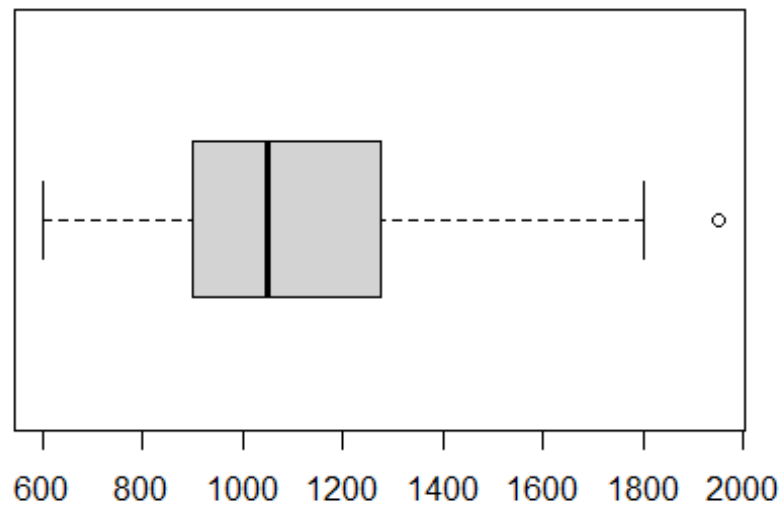
## Registered S3 methods overwritten by 'car':
##   method             from
##   na.action.merMod    lme4
##   hist.boot           FSA
##   confint.boot        FSA

##
## Attaching package: 'car'
```

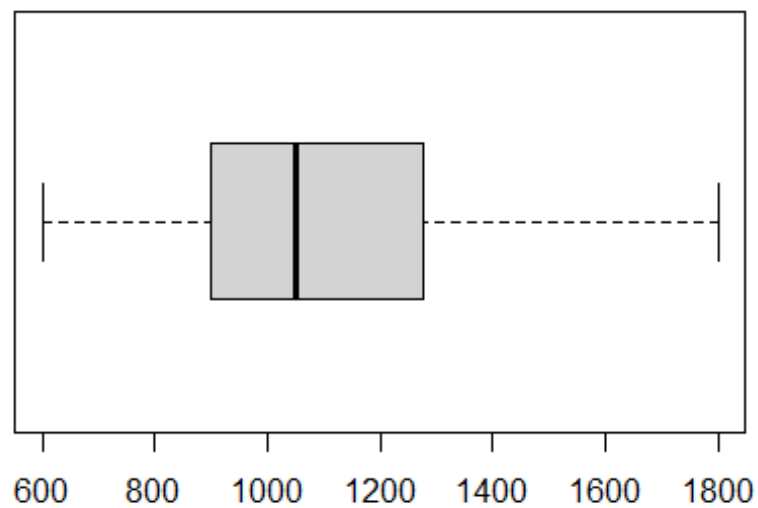
```
## The following object is masked from 'package:DescTools':  
##  
##      Recode  
  
## The following objects are masked from 'package:mosaic':  
##  
##      deltaMethod, logit  
  
## The following object is masked from 'package:dplyr':  
##  
##      recode  
  
## The following object is masked from 'package:FSA':  
##  
##      bootCase  
  
## The following object is masked from 'package:psych':  
##  
##      logit  
  
library(carData)  
library(vioplot)  
  
## Loading required package: sm  
## Package 'sm', version 2.2-6.0: type help(sm) for summary information  
## Loading required package: zoo  
  
##  
## Attaching package: 'zoo'  
  
## The following objects are masked from 'package:base':  
##  
##      as.Date, as.Date.numeric
```

krabicove grafy

```
boxplot(plat, horizontal=T)#graficky
```



```
boxplot(plat, horizontal=T, outline=F)
```

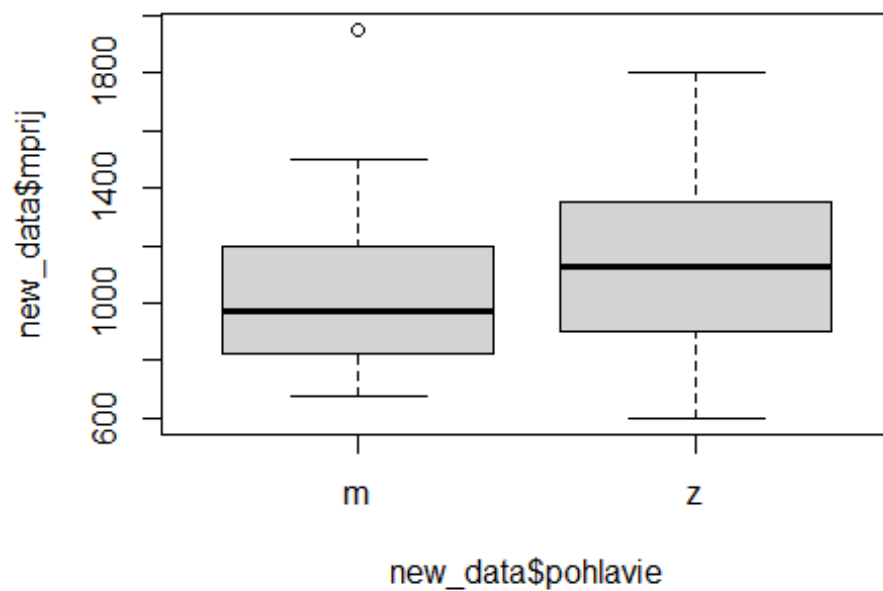


```
boxplot.stats(plat)
```

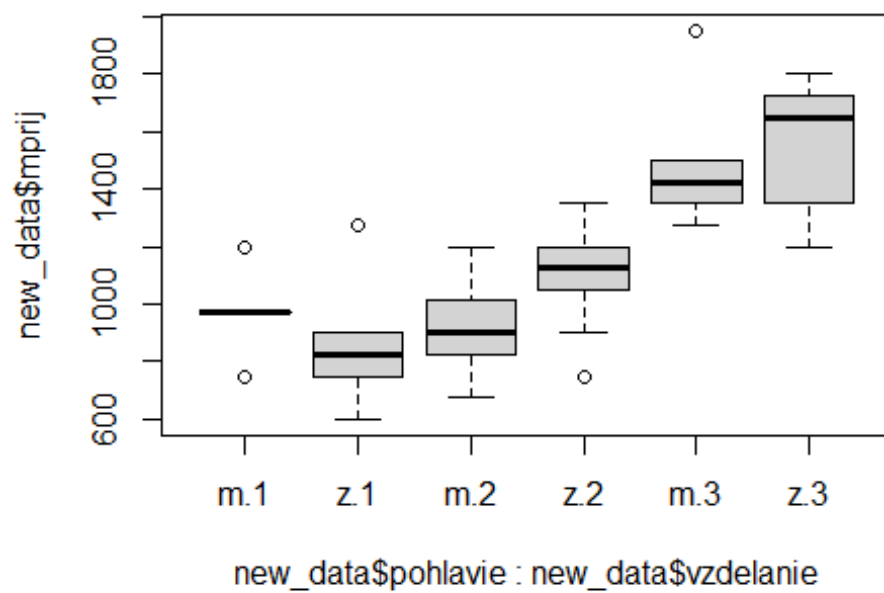
```
## $stats
## [1] 600 900 1050 1275 1800
##
## $n
## [1] 45
##
## $conf
## [1] 961.6753 1138.3247
##
## $out
## [1] 1950
```

pre podmnožiny

```
boxplot(new_data$mprij~new_data$pohlavie)
```

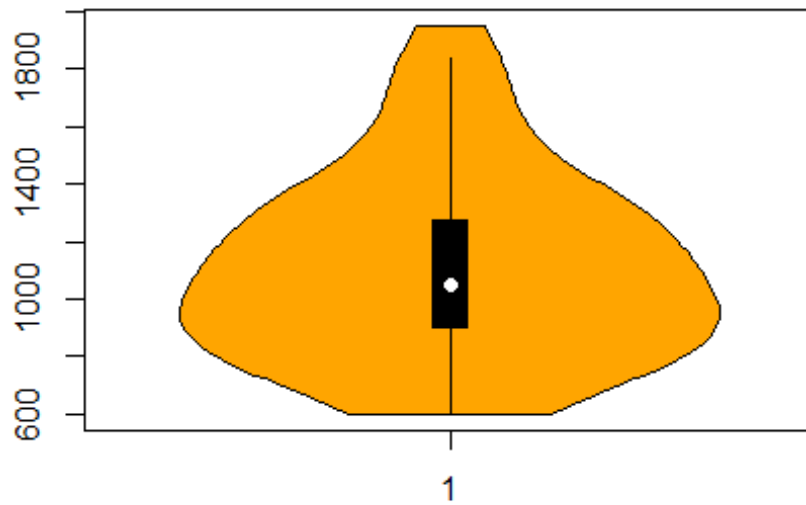


```
boxplot(new_data$mprij~new_data$pohlavie+new_data$vzdelanie)
```



Huslovy graf

```
vioplot(plat,col="orange")
```



```
par(mfrow=c(1,2))  
boxplot(plat,col="blue")  
vioplot(plat,col="orange")
```

